

Parking Lot System Architecture

Patterns: Factory Pattern, Strategy Pattern (Spot Allocation & Fee), Singleton Pattern | Domain: Resource Booking / Strategy Pattern

Core Requirements & Features	Core Domain Classes & Interfaces
• Multi-Vehicle Support: Park Motorcycle, Car, Bus in matching spot types. • Multi-Level Parking: Manage floors, entry/exit gates, and spot availability. • Ticket & Fee Computation: Issue ticket at entry, calculate fee based on duration & vehicle type.	• ParkingLot: Singleton managing floors, gates, & global capacity. • ParkingSpot: Base class for Compact, Large, & Handicapped spots. • SpotAllocationStrategy: Nearest-to-Entrance or First-Available allocation strategy. • FeeCalculationStrategy: Flat-rate, Hourly, or Vehicle-type weighted fee logic.

Critical Execution Workflow & Method Trace

1. Vehicle arrives at Entry Gate 1.
2. Entry Gate queries SpotAllocationStrategy.findSpot(vehicleType).
3. Strategy iterates floors, scanning available spots matching vehicle dimensions.
4. Marks selected ParkingSpot as OCCUPIED (isAvailable = false).
5. Generates ParkingTicket (ticketId, vehicleNumber, spotId, entryTime).
6. Vehicle departs at Exit Gate 2: Gate passes ParkingTicket to FeeCalculationStrategy.
7. Fee strategy calculates duration = exitTime - entryTime and applies vehicle hourly multiplier.
8. Payment processed; ParkingSpot marked AVAILABLE (isAvailable = true), notifying display boards.

Concurrency & Thread Safety Mechanics	Design Trade-offs & Interview FAQs
• Spot Race Condition: Atomic boolean compareAndSet on ParkingSpot.isAvailable. • Gate Synchronization: Thread-safe ConcurrentHashMap for active tickets across multiple gates. • Display Board: Volatile counter for real-time available spot display updates.	• Fixed vs Flexible Spots: 1 Bus taking 5 Car spots vs dedicated Bus spots; dedicated spots simplify locking. • Strategy Decoupling: FeeCalculationStrategy can be swapped without modifying ParkingTicket or Gate code. • Single-Instance: Singleton ParkingLot simplifies state; requires distributed cache for multi-building lots.